

# Studying Esports Competition: Piloting Methodology for User Studies During Tournaments

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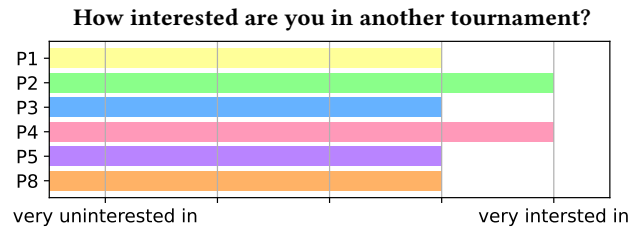


Figure 1: (Left) A picture taken shortly before the Rocket League Tournament at SIGGRAPH 2023 Labs venue. (Right) Participants were asked about interest in future tournaments and all replied positively.

## CCS CONCEPTS

• Human-centered computing → User studies; • Applied computing → Computer games.

## KEYWORDS

tournaments, experiments, competitive play, rocket league

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## 1 INTRODUCTION

Esports impacts society by creating jobs [Johnson and Woodcock 2021], hardware innovation [Li et al. 2022], and more. It also provides new opportunities to study mastery and expertise when using computers [Boudaoud et al. 2022]. Not surprisingly, esports research is growing. But, as in many emerging research fields, there are challenges. One is the disconnect between competition, fundamental to esports and indeed all sports, and traditional experimentation. A lab environment allows for tight control of experimental variables, but does not easily accommodate competition.

To meet such challenges, we suggest tournament-experiment hybrids, in which participants are also competitors. As a case study of

this approach, we held an eight-player Rocket League tournament-experiment at SIGGRAPH 2023. The experiment studied the performance trade-off between how good the game looks and how quickly it renders. In this talk, we offer a preliminary assessment of the viability of experimentation during tournaments, and of tournament play with embedded experimentation.

## 2 BACKGROUND

Designing experiments is a well-studied, complex task with many conflicting constraints. In experimental user studies, the overarching goal is to gain understanding of how a system affects users by manipulating independent variables (e.g., an interface's configuration) while measuring dependent variables (e.g., the user's performance). Achieving this goal requires significant compromise between many conflicting sub-goals including *internal validity*, that the effects on dependent variables were caused by manipulations of independent variables, not other uncontrolled variables; *external validity*, that the findings are generalizable enough to have meaning outside the lab; and *feasibility*, that the experiment can be performed without unreasonable time and cost.

Tournament design is also quite complex with conflicting constraints, but is less well understood by user study researchers. Both Csató [Csató 2021] and Placek [Placek 2023] identify fairness (whether the design advantages one player over another) and fun (whether the design entertains) as primary goals. These and other goals result in a number of different designs commonly seen in sports as well as esports with the Counter-Strike: Global Offensive championships using single-elimination, the League of Legends (LoL) and DOTA 2 championships using double-elimination, and LoL using round-robin for play-in rounds.

Tournaments generate competitive pressures that have significant effects on performance. Positive effects include more intrinsic motivation [Tauer and Harackiewicz 2004] and benchmarking comparisons to competitors [Rhea et al. 2003], while negative effects

include anxiety leading to choking [Beilock and Carr 2001] and avoidance [Thomassen and Halvari 1996]. One concurrent study attempted a combined tournament as a part of an experiment, but in a first-person shooter game [Tokey et al. 2024].

### 3 A PILOT TOURNAMENT-EXPERIMENT

To begin learning about how to maximize the advantages of hybrid tournament-experiments while minimizing their challenges, we piloted an experimental study of visual and temporal detail effects on gameplay by holding an esports tournament. Our research questions were twofold: experimentally, do visual/temporal game setting choices affect game performance? Methodologically, does introducing experimentation into tournament play affect perceived fun and fairness? We performed this experiment at the ACM SIGGRAPH 2023 Conference in Los Angeles, in the Labs venue. Participants had to be registered SIGGRAPH attendees. We recruited participants through small ads on the SIGGRAPH webpage, and during a presentation at a SIGGRAPH esports workshop.

We chose Rocket League by Epic Games (patch v2.31) for this tournament-experiment, because the short match time fit well into the two-hour period scheduled for us by conference organizers. Rocket League is a well-known esports title with nearly 100 million monthly players. We used the official Rocket League tournament tool for the event. Players chose their preferred input method between mouse and keyboard or controller/joystick.

We used a simple, single-factor experimental design. The independent variable was *Settings*, which had two levels: *high* (*H*) and *low* (*L*). Our tournament design was also simple, using a 3-round, 8-player single-elimination bracket. In each match, players competed in at least two games, once with the *H* and again with the *L* levels, making *Settings* a within-participants variable.

To balance tournament fairness and experimental control, each two-player match was best-of-three, where the first two games had alternating *H/L* and *L/H* treatments for the two player-participants. If needed, the third game in any match was randomly chosen to be either *H/H* or *L/L* in an effort to preserve match fairness.

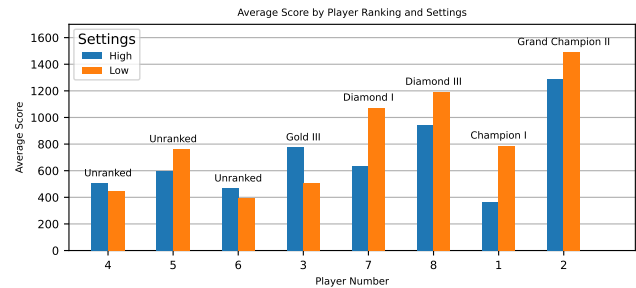
### 4 RESULTS

We collected two types of results from the tournament-experiment pilot. First, we conducted a subjective survey of participants focusing on fairness, fun, and the effects of experimental game settings. One example from this survey is found in Figure 1 (right). Overall, participants found the tournament fun and fair, and did not think that experimental changes in settings had much effect.

Second, we measured player performance, primarily as goals scored and their tournament win/loss record. Score (an in-game metric) is depicted in Figure 2. The mean score in games with the *L* settings was 950 ( $se=140$ ), while with the *H* settings it was 813 ( $se=104$ ). Most players achieved higher scores with *L* settings, especially higher-ranked players. We found similar trends with winning percentage: the mean winning percentage with the *L* settings was 57 ( $se=14$ ), while with the *H* settings it was 43 ( $se=13$ ).

### 5 CONCLUSIONS

Overall, the results of our pilot tournament-experiment are encouraging. Experimentally, despite (and perhaps because of) the



**Figure 2: The average scores for each player across all rounds they played, grouped by detail Settings. The players are sorted by their rankings, with low/unranked players to the left and highly skilled players to the right.**

tournament, we found trends indicating that low detail settings (which reduced latency and increased frame rate) improved performance, particularly among more skilled players. There is a chance that in a more extensive experiment, these trends may be statistically significant, rewarding investigation of esports in its “natural habitat”. Yet the methodological results are still more interesting. Participants felt that the tournament was fun and fair despite experimental intervention, and were eager to participate again.

As experimenters, we found it much easier to schedule these participant/players, with no one raising concerns about calendar conflicts. On the other hand, these experiments moved quickly, requiring three investigators to keep even this small, eight-person tournament moving along. In the future we will conduct additional studies to further address various open questions.

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